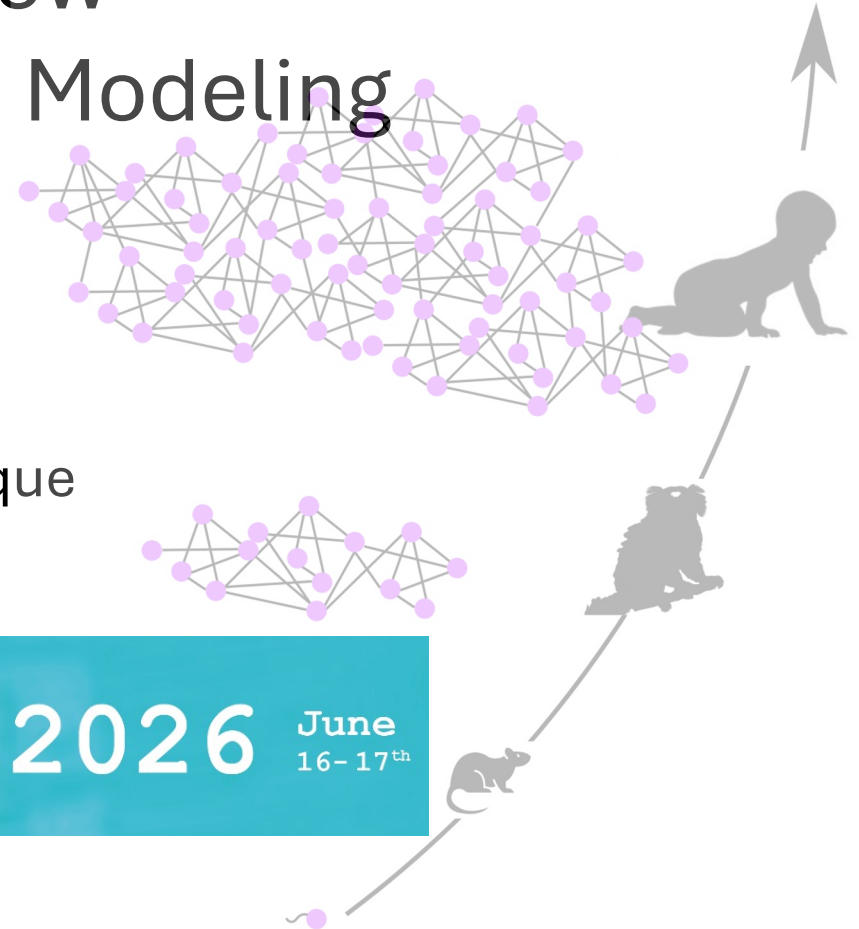




Neuro-Workflow Agent-Assisted Brain Modeling

Gutierrez, Carlos Enrique
OIST



About Brain/MINDS 2.0

Multidisciplinary Frontier Brain and Neuroscience Discoveries (Brain/MINDS 2.0)

[Prioritized Research Topics]

Development of therapeutic methods and new drug candidates

- Identification of **biomarkers** for ultra-early diagnosis
- Identification of novel therapeutic targets other than amyloid-beta ($A\beta$) in Alzheimer's disease, e.g.

Translational/Reverse Translational collaboration between basic and clinical research

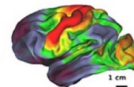
Pathophysiology of brain diseases

- Elucidation of the pathophysiological mechanisms of neurological and psychiatric disorders
- Development of disease model animals

Promotion of industry-academia collaboration and international partnerships

Elucidation of human brain function and dynamics

- Understanding the dynamics of neural circuits and other aspects of the human brain
- Multidimensional and multilevel data analysis, etc.



- ※1 Development of a "Digital Brain"
- Recreating the dynamics of higher brain functions in a digital space

Development of a "Digital Brain"※1

Establishment of innovative technologies and research infrastructure

- Establishment of infrastructure for human fMRI imaging and animal resources
- Development of mathematical models, brain measurement technologies, and related applications

Contributing to

Groundbreaking diagnostics, treatments, and drug discovery for diseases such as dementia, while also expanding a wide range of industrial applications

Software

We are developing **Neuro-Workflow**, an AI-assisted GUI tool for neural data analysis, model building, simulation, visualization, and optimization.

<https://digitalbrainproject.github.io/home/>



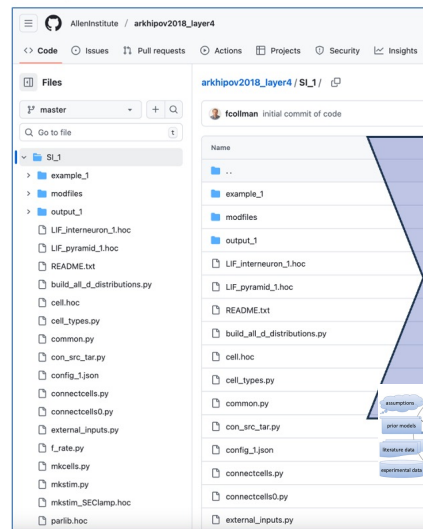
The Brain/MINDS 2.0 Neuro-Workflow



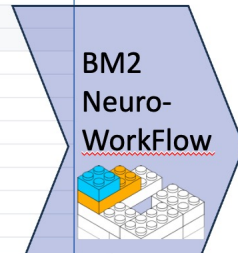
Neuro-Workflow is a node-based framework that represents brain models as self-describing graphs, making it easy for researchers, non-experts, and AI agents to assemble, reuse, and extend them. It aims to serve as the BM2 project's shared model repository.

Models are distributed as a collection of unstructured files.

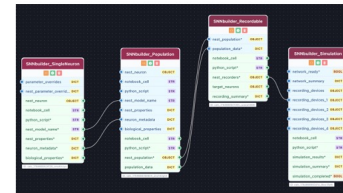
e.g., Layer IV of mouse V1 [Allen Institute; Arkhipov et al., 2018]



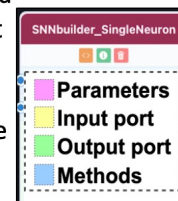
This makes the model difficult to understand, experiment with, reuse, and extend.



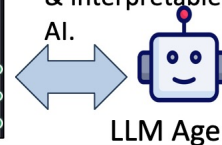
Models are represented as workflows (Nodes & Edges)



Nodes use a schema that makes models easier to use

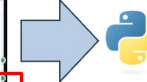
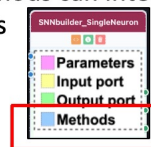


The schema makes nodes discoverable & interpretable by AI.



LLM Agents

Each node is a Python class; its methods can interface with popular tools



TVB, NEST, BMTK, BMCR, Custom code, etc.

Nodes let you build new models and integrate existing models.



Neuro-Workflow demos

Marmoset full brain model simulation

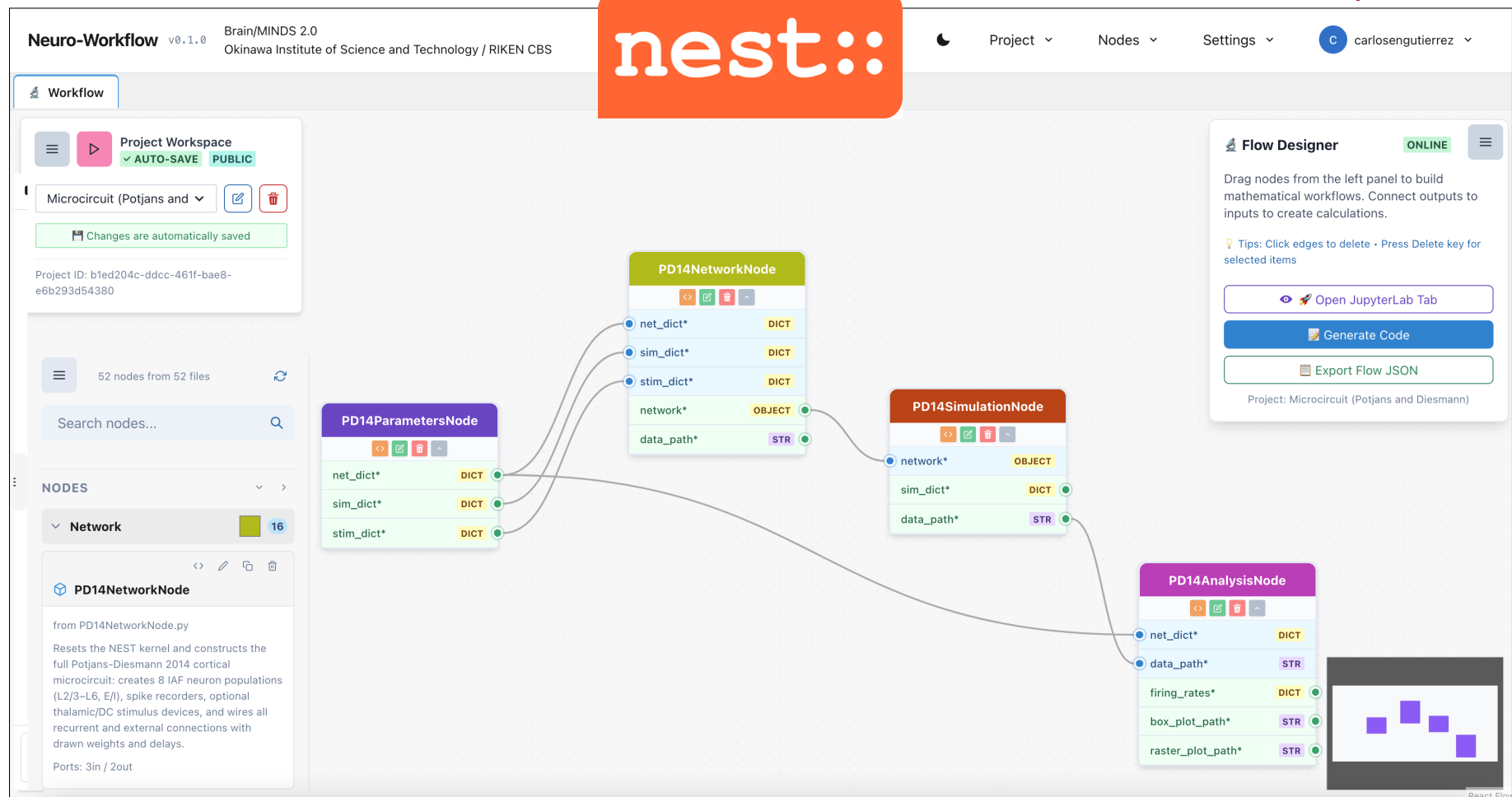
https://youtu.be/_FAjMHKHhGw?si=LkZTVnfoe9YHC2nE

Basal Ganglia Model of the macaque

<https://youtu.be/HvcTYz3RIM8?si=NvDpF5-Kb6qy0tfO>



The Brain/MINDS 2.0 Neuro-Workflow



Model Builder



Single Neuron simulation:

NW_IClamp → NW_Population → NW_SimConfig → NW_Analysis

```
from neuroworkflow import WorkflowBuilder
from neuroworkflow.nodes.sandbox.NW_Population import NW_Population
from neuroworkflow.nodes.sandbox.NW_SimConfig import NW_SimConfig
from neuroworkflow.nodes.sandbox.NW_Analysis import NW_Analysis
from neuroworkflow.nodes.sandbox.NW_IClamp import NW_IClamp
```

2. Configure nodes

```
pop.configure(
    pop_name      = "v1",
    N             = 1,
    model_type    = "point_neuron",
    model_template = "nest:iaf_psc_alpha",
    ei_type       = "exc",
    location      = "VISp",
    layer         = "L4",
    nest_params = {
        "C_m": 250.0, # pF membrane capacitance
        "tau_m": 10.0, # ms membrane time constant
        "t_ref": 2.0, # ms refractory period
        "V_th": -55.0, # mV spike threshold
        "V_reset": -70.0, # mV reset potential
        "E_L": -70.0, # mV resting potential
    },
)

clamp.configure(
    amp_na = 0.150,
    delay_ms = 500.0,
```

1. Instantiate nodes

```
: pop      = NW_Population("Population")
   cfg     = NW_SimConfig("SimConfig")
   ana     = NW_Analysis("Analysis")
   clamp   = NW_IClamp("IClamp")
```

3. Build and run the workflow

```
wf = WorkflowBuilder("NW_SingleCell_PointNeuron")
wf.add_node(clamp)
wf.add_node(pop)
wf.add_node(cfg)
wf.add_node(ana)

wf.connect("IClamp", "iclamp", "Population", "iclamp")
wf.connect("Population", "population", "SimConfig", "populations")
wf.connect("SimConfig", "results", "Analysis", "results")

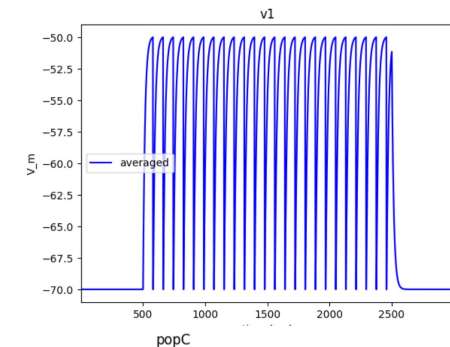
wf.context["results_path"] = "./results/run1"

workflow = wf.build()
ok = workflow.execute()
assert ok, "Workflow failed – check printed errors above"
print("Workflow completed successfully")
```


Model Builder

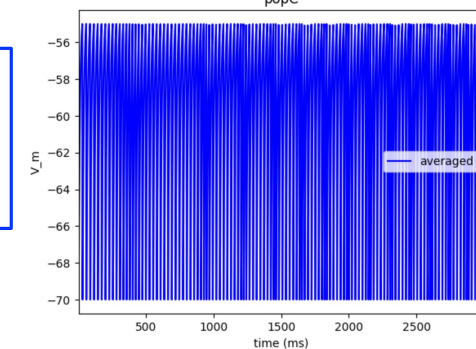
Single Neuron simulation:

NW_IClamp → NW_Population → NW_SimConfig → NW_Analysis



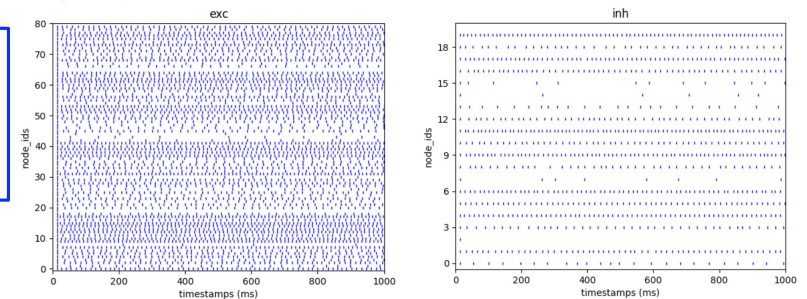
Ring Network - 3 Point Neurons

PopA }
PopB } → NW_Connectivity (ring matrix) → NW_SimConfig → NW_Analysis
PopC }

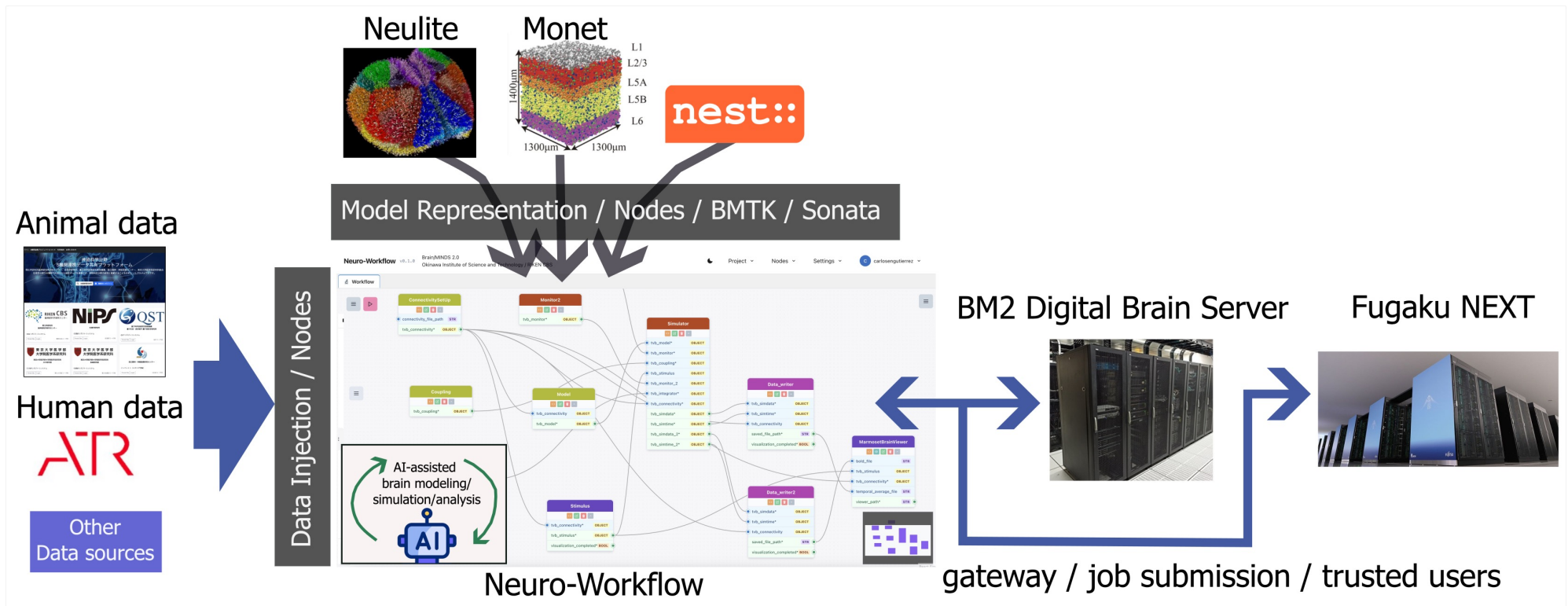


Balanced E-I Network

PopExc }
PopInh } → NW_Connectivity → NW_SimConfig → NW_Analysis



Neuro-Workflow: Models and Data Integration



Conclusions

Neuro-Workflow is built for integration with AI

Interoperability focus

Moving towards Multi-agent framework (AI4Science)

Community building is critical (user base driving innovation)

Openness